



Linear Models III: Categorical Predictors

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Stat 141 | Fall 2026
Monday, Week 5

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Goals for Today

- Recap: Simple linear regression model
- Broadening our idea of linear regression
- Regression with a single, binary categorical explanatory variable
- Regression with a single categorical explanatory variable with more than 2 levels

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Simple Linear Regression

So far we've considered this model when:

- Response variable (y): quantitative
- Explanatory variable (x): quantitative
 - Have only ONE explanatory variable.
- AND, $f()$ can be approximated by a line:

$$y = \beta_0 + \beta_1 x + \epsilon$$

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Linear Regression

Linear regression is a flexible class of models that allow for:

- Both quantitative and categorical **explanatory** variables.
- **Multiple** explanatory variables.
- A **quantitative response variable**.

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Exploratory Data Analysis

Before building the model, let's explore and visualize the data!

- Q: What **dplyr** functions should we use to find the mean and sd of **winpercent** by the categories of **chocolate**?
- Q: What graph should we use to visualize the **winpercent** scores by **chocolate**?

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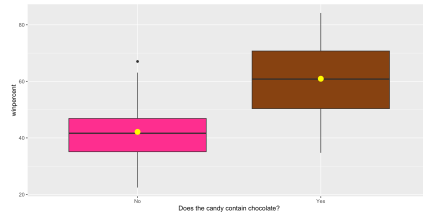
Exploratory Data Analysis

```
1 # Summarize
2 candy %>%
3   group_by(chocolate) %>%
4   summarize(count = n(),
5             mean_win = mean(winpercent),
6             sd_win = sd(winpercent))
# A tibble: 2 x 4
  chocolate count mean_win sd_win
  <dbl> <int> <dbl> <dbl>
1         0    48    42.1   10.2
2         1    37    60.9   12.8
```

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Exploratory Data Analysis

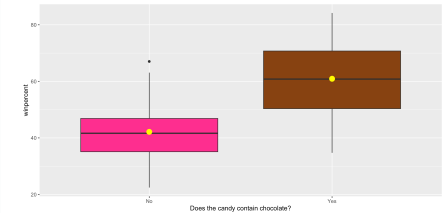
```
1 ggplot(candy, aes(x = factor(chocolate),
2                     y = winpercent,
3                     fill = factor(chocolate))) +
4   geom_boxplot() +
5   stat_summary(fun = mean,
6               geom = "point",
7               color = "yellow",
8               size = 4) +
9   guides(fill = "none") +
10  scale_fill_manual(values =
11                    c("0" = "deeppink",
12                      "1" = "chocolate4")) +
13  scale_x_discrete(labels = c("No", "Yes"),
14                  name =
15    "Does the candy contain chocolate?")
```



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Exploratory Data Analysis

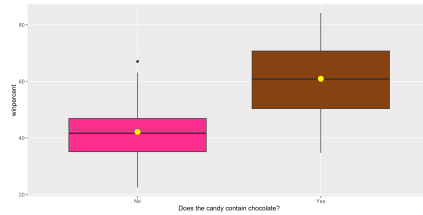
```
1 ggplot(candy, aes(x = factor(chocolate),
2                     y = winpercent,
3                     fill = factor(chocolate))) +
4   geom_boxplot() +
5   stat_summary(fun = mean,
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Exploratory Data Analysis

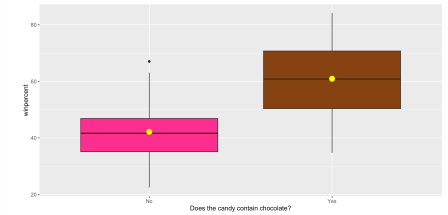
```
1 ggplot(candy, aes(x = factor(chocolate),
2                     y = winpercent,
3                     fill = factor(chocolate))) +
4   geom_boxplot() +
5   stat_summary(fun = mean,
6               geom = "point",
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Exploratory Data Analysis

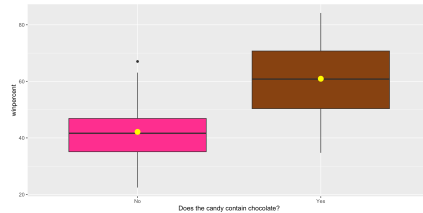
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15                    "Does the candy contain chocolate?")
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Exploratory Data Analysis

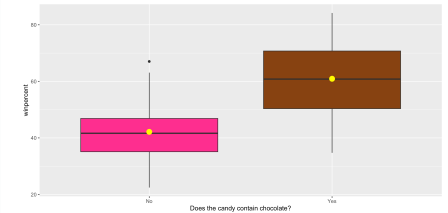
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3                     fill = factor(chocolate))) +
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5   stat_summary(fun = mean,
6               geom = "point",
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Exploratory Data Analysis

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```



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Fit the Linear Regression Model

Model Form:

$$y = \beta_0 + \beta_1 x + \epsilon$$

```
1 mod <- lm(winpercent ~ chocolate, data = candy)
2 library(moderndive)
3 get_regression_table(mod)

# A tibble: 2 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 42.1        1.65      25.6     0      38.9    45.4
2 chocolate 18.8        2.50       7.52     0      13.8    23.7
```

Q: When $x = 0$, what is $\hat{y}$?

Q: When $x = 1$, what is $\hat{y}$?

Different Interpretation for Categorical Explanatory Variables

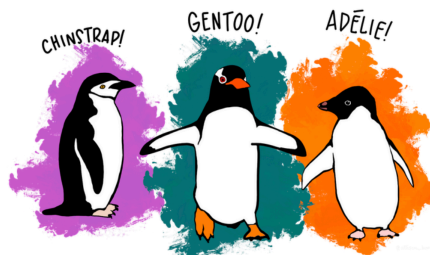
- When the explanatory variable is categorical, β_0 and β_1 no longer represent the intercept and slope.
- Now β_0 represents the (population) mean of the response variable when $x = 0$.
- β_1 represents the change in the (population) mean response going from $x = 0$ to $x = 1$.
- Can still do prediction (Q: how many unique predicted values will we get?)

```
1 new_candy <- data.frame(chocolate = c(0, 1))
2 predict(mod, newdata = new_candy)

      1      2
42.14226 60.92153
```

New example: Palmer Penguins

```
1 library(palmerpenguins)
```



The Palmer Archipelago penguins. Artwork by @allison_horst.

Take a look at the data

```
1 glimpse(penguins)

Rows: 344
Columns: 8
$ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel...
$ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgersen...
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ...
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ...
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186...
$ body_mass_g   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ...
$ sex          <fct> male, female, female, NA, female, male, female, male...
$ year         <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007...
```

We'd like to predict a penguin's bill length based on their species.

Response variable?

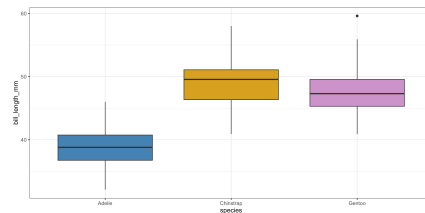
Explanatory variable?

Exploratory data analysis

```
1 penguins %>%
2   group_by(species) %>%
3   summarize(
4     avg_bill_length = mean(bill_length_mm,
5                           na.rm = TRUE)
6   )
```

```
# A tibble: 3 × 2
  species avg_bill_length
  <fct>    <dbl>
1 Adelie      38.8
2 Chinstrap  48.8
3 Gentoo     47.5
```

```
1 ggplot(penguins,
2       aes(x = species,
3           y = bill_length_mm,
4           fill = species)) +
5   geom_boxplot() +
6   scale_fill_manual(values = c("steelblue",
7                                "goldenrod",
8                                "plum3")) +
9   guides(fill = "none") +
10  theme_bw()
```



How do we handle more than 2 groups???

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Boardwork

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Fit the model in R

$$y = \beta_0 + \beta_1 x_{\text{species:Chinstrap}} + \beta_2 x_{\text{species:Gentoo}} + \epsilon$$

R automatically makes species indicators for us and chooses a **reference level**.

```
1 penguin_mod <- lm(bill_length_mm ~ species, penguins)
2 get_regression_table(penguin_mod)

# A tibble: 3 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept    38.8      0.241     161.    0      38.3    39.3
2 species: Chinstrap 10.0      0.432     23.2    0       9.19   10.9
3 species: Gentoo   8.71      0.36      24.2    0       8.01    9.42
```

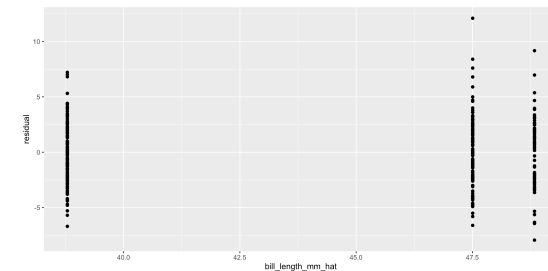
• Q: What is the equation for the fitted model?

$$\begin{aligned} \hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 \cdot x_{\text{species:Chinstrap}} + \hat{\beta}_2 \cdot x_{\text{species:Gentoo}} \\ &= 38.8 + 10.0 \cdot x_{\text{species:Chinstrap}} + 8.71 \cdot x_{\text{species:Gentoo}} \end{aligned}$$

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Remember to diagnose your models!

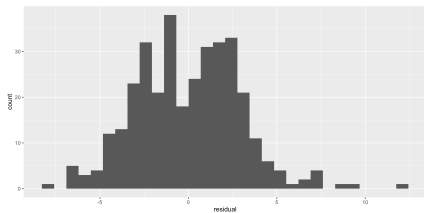
```
1 library(modernr)
2 res <- get_regression_points(penguin_mod)
3 ggplot(res, aes(x = bill_length_mm_hat, y = residual)) +
4   geom_point()
```



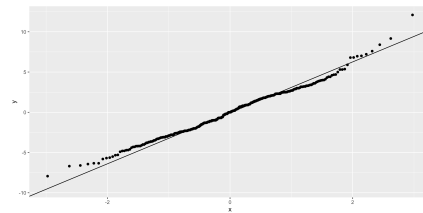
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Remember to diagnose your models!

```
1 ggplot(res, aes(x = residual)) +
2   geom_histogram()
```



```
1 ggplot(res, aes(sample = residual)) +
2   geom_qq() +
3   geom_qq_line()
```



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Multiple Linear Regression: A peak ahead

Recall our penguin model

$$y = \beta_0 + \beta_1 x_{\text{species:Chinstrap}} + \beta_2 x_{\text{species:Gentoo}} + \epsilon$$

Even though we are using one predictor (species), we now have β_0 , β_1 , and β_2 !

- We **recoded** the species predictor into two binary predictors
- We are actually doing **multiple** linear regression now
- **Later this week:** We'll formalize and extend multiple linear regression
- **Categorical explanatory variables are tricky!!** Revisit these examples, more practice with HW 5 Exercise 1

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Bonus Activity: Changing Reference Level

We can change the **reference level** if we want.

```
1 penguins$species <- relevel(penguins$species, ref = "Chinstrap")
2 penguin_mod <- lm(bill_length_mm ~ species, penguins)
3 get_regression_table(penguin_mod)
```

```
# A tibble: 3 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>    <dbl>    <dbl>   <dbl>    <dbl>    <dbl>
1 intercept  48.8      0.359    136.     0      48.1     49.5
2 species: Adeline -10.0     0.432   -23.2    0      -10.9    -9.19
3 species: Gentoo -1.33     0.447    -2.97   0.003    -2.21   -0.449
```

Get in groups of ~5 and find a spot on one of the boards.

- **Q1:** Write down the equation for this fitted model.
- **Q2:** Draw a few rows of the data frame with the variables used by the model.
- **Q3:** How is the interpretation of $\hat{\beta}_0$ different now?
- **Q4:** How should we interpret the other two coefficients?

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Bonus Activity: Changing Reference Level (Answers)

- **Q1:** Write down the equation for this fitted model.

$$\hat{Y} = 48.8 - 10.0 x_{\text{species:Adeline}} - 1.3 x_{\text{species:Gentoo}}$$

- **Q2:** Draw a few rows of the data frame with the variables used by the model.

```
# A tibble: 6 x 4
# Groups:   species [3]
  bill_length_mm species x_adeline x_gentoo
<dbl> <fct>    <dbl>    <dbl>
1    49.8 Chinstrap      0        0
2    52 Chinstrap      0        0
3    38.6 Adeline        1        0
4    43.1 Adeline        1        0
5    48.5 Gentoo         0        1
6    42.7 Gentoo         0        1
```

- **Q3:** How is the interpretation of $\hat{\beta}_0$ different now?
 - β_0 ($\hat{\beta}_0$) now represents (estimates) the population mean bill length for Chinstrap penguins, not Adeline penguins
- **Q4:** How should we interpret the other two coefficients?
 - They now represent the predicted change in bill lengths for Adeline and Gentoo penguins on average, relative to the predicted bill length for Chinstrap penguins.

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